

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

The speed of a vehicle is increasing at a rate of **7.3** meters per second squared. What is this rate, in **miles per minute squared**, rounded to the nearest tenth? (Use **1 mile = 1,609 meters**.)

Answer

- A. 0.3
- B. 16.3
- C. 195.8
- D. 220.4

Correct Answer: B

Rationale

Choice B is correct. It's given that the speed of a vehicle is increasing at a rate of **7.3** meters per second squared. It's given to use **1 mile = 1,609 meters**. There are **60** seconds in **1** minute; therefore, **60²** or **3,600** seconds squared is equal to **1** minute squared. It follows that the rate of **7.3** meters per second squared is equivalent to $\left(\frac{7.3 \text{ meters}}{1 \text{ second squared}}\right)\left(\frac{1 \text{ mile}}{1,609 \text{ meters}}\right)\left(\frac{3,600 \text{ seconds squared}}{1 \text{ minute squared}}\right)$, or approximately **16.33 miles per minute squared**. The rate, in **miles per minute squared**, rounded to the nearest tenth is **16.3**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.